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No.
Date :

2. Dik : $E = A \cdot P$
$P = A^{-1} \cdot E$

$$A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix} \quad a = 1 \quad b = 1 \quad N = 11$$

Dit :

Jwb :

a. tulis matriks A (a dan b), Hitung det(A)

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix}$$

$$\det(A) = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix} = 2\begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} = 2(12-1) = 22$$

$$\begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix} = 1\begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix} = 1(4-1) = 3$$

$$\begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix} = 2\begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix} = 2(-2) = -4$$

$\det(A) = 22 - 3 - (-4) = 15$ //

b. hitung Matriks Minor M_{ij} $i, j \in \{1, 2, 3\}$. total 9 minor

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix}$$

$$M_{11} = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} = 12 - 1 = 11$$

$$M_{12} = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix} = 4 - 1 = 3$$

$$M_{13} = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix} = 1 - 3 = -2$$

$$M_{21} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix} = 4 - 2 = 2$$

$$M_{22} = \begin{bmatrix} 2 & 2 \\ 1 & 4 \end{bmatrix} = 8 - 2 = 6$$

$$M_{23} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = 2 - 1 = 1$$

$$M_{31} = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} = 1 - 6 = -5$$

$$M_{32} = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} = 2 - 2 = 0$$

$$M_{33} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = 6 - 1 = 5$$

$$= \begin{bmatrix} 11 & 3 & -2 \\ 2 & 6 & 1 \\ -5 & 0 & 5 \end{bmatrix}$$ //

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c. Matriks kofaktor C, matriks Adjoin $Adj(A) = C^T$, Hitung $A^{-1} = (1/\det(A)) \cdot Adj(A)$

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix} \qquad \det(A) = 15 \qquad M = \begin{bmatrix} 11 & 3 & -2 \\ 2 & 6 & 1 \\ -5 & 0 & 5 \end{bmatrix}$$

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \rightarrow C = \begin{bmatrix} 11 & -3 & -2 \\ -2 & 6 & -1 \\ -5 & 0 & 5 \end{bmatrix} //$$

$$Adj(A) = C^T = \begin{bmatrix} 11 & -2 & -5 \\ -3 & 6 & 0 \\ -2 & -1 & 5 \end{bmatrix} //$$

$$A^{-1} = \frac{1}{\det(A)} \cdot Adj(A$$

$$= \frac{1}{15} \cdot \begin{bmatrix} 11 & -2 & -5 \\ -3 & 6 & 0 \\ -2 & -1 & 5 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{11}{15} & -\frac{2}{15} & -\frac{5}{15} \\ -\frac{3}{15} & \frac{6}{15} & 0 \\ \frac{-2}{15} & -\frac{1}{15} & \frac{5}{15} \end{bmatrix} = \begin{bmatrix} \frac{11}{15} & -\frac{2}{15} & -\frac{1}{3} \\ -\frac{1}{5} & \frac{2}{5} & 0 \\ -\frac{2}{15} & -\frac{1}{15} & \frac{1}{3} \end{bmatrix} //$$

e. $$P = \begin{bmatrix} 120 & (50+N) & 200 \\ 80 & 1+2 & (10+N) \\ (30+N) & 90 & 170 \end{bmatrix}$$

enkripsi $E = A \cdot P$

dekripsi $P' = A^{-1} \cdot E$

Buktikan $P' = P$

$$P = \begin{bmatrix} 120 & (50+11) & 200 \\ 80 & (1+2) & (10+11) \\ (30+11) & 90 & 170 \end{bmatrix} = \begin{bmatrix} 120 & 61 & 200 \\ 80 & 3 & 21 \\ 41 & 90 & 170 \end{bmatrix}$$

$$E = A \cdot P$$

$$= \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 120 & 61 & 200 \\ 80 & 3 & 21 \\ 41 & 90 & 170 \end{bmatrix}$$

Baris kolom

11 $= 2(120) + 1(80) + 2(41) = 402$

12 $= 2(61) + 1(3) + 2(90) = 305$

13 $= 2(200) + 1(21) + 2(170) = 761$

21 $= 1(120) + 3(80) + 1(41) = 401$

22 $= 1(61) + 3(3) + 1(90) = 160$

23 $= 1(200) + 3(21) + 1(170) = 433$

31 $= 1(120) + 1(80) + 4(41) = 364$

32 $= 1(61) + 1(3) + 4(90) = 424$

33 $= 1(200) + 1(21) + 4(170) = 901$

$$E = \begin{bmatrix} 402 & 305 & 761 \\ 401 & 160 & 433 \\ 364 & 424 & 901 \end{bmatrix}$$

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$$P' = A^{-1} - E$$
$$= A^{-1}(AP)$$
$$= (A^{-1}A)P$$
$$= IP$$
$$= P$$

$$P' = \begin{bmatrix} 120 & 61 & 200 \\ 80 & 3 & 21 \\ 41 & 90 & 170 \end{bmatrix}$$

$P' = P$ //

Jika $\det(A) = 0$
maka matriks A tidak memiliki invers
sehingga proses dekripsi tidak bisa dilakukan.

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